

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
from google.colab import files
uploaded=files.upload()
```

No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving titanic .csv to titanic (1).csv

```
df = pd.read_csv("titanic .csv")
```

```
df
```

```
df.head()
```

	PassengerId	Survived	Class	Name	Sex	Age	SibSp	Spa	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	0	A/5 21171	7.2500	NaN	
11	2 2	1 1	1 1	Cummings, Mrs. John Bradley (Florence Briggs Th. ...)	female	38.0	1 1	0 0	0	PC 17599	71.2833	C86	5
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	0	STON/O2. 3101282	7.9250	NaN	
33	4 4	1 1	1 1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1 1	0 0	1 1	113803	53.1000	C123	23
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	0	373450	8.0500	NaN	

```
df.shape
```

(891, 12)	887	0	2	Rev.	male	27.0	0	0	211536	13.0000	NaN
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```
df.info()
```

```

887      <class 'pandas.core.frame.DataFrame'> Margaret
888      RangeIndex: 891 entries, 0 to 890      Edith
889      Data columns (total 12 columns):
890      #      Column      Non-Null Count  Dtype
891      #      :      #      :              #      :
892      0      name      891            object
893      1      sex       891            object
894      2      age       891            int64
895      3      sibship   891            int64
896      4      parch     891            int64
897      5      ticket    891            object
898      6      fare      891            float64
899      7      cabin     294            object
900      8      embarked  891            object
901      9      class     891            object
902      10     status    891            object
903      11     name      891            object
904      12     name      891            object
905      13     name      891            object
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1113     221    name      8
```

```

888 ----- 889 ----- 0 ----- 3 --- Catherine female NaN 1 2 6607 23.4500 NaN
0 PassengerId 891 non-null int64
1 Survived 891 non-null int64
2 Pclass 891 non-null int64
3 Name 891 non-null object
489 Sex 890 891 non-null object Karl male 26.0 0 0 111369 30.0000 C148
5 Age 714 non-null float64
6 SibSp 891 non-null int64
7 Parch 891 non-null int64
890 Ticket 891 891 non-null object Mr. male 32.0 0 0 370376 7.7500 NaN
9 Fare 891 non-null float64
10 Cabin 204 non-null object
11 Embarked 889 non-null object
891 rows x 12 columns
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB

```

```
df.duplicated().value_counts()
```

	count
False	891

```
dtype: int64
```

```

# Check how many missing values exist in each column
print(df.isnull().sum())

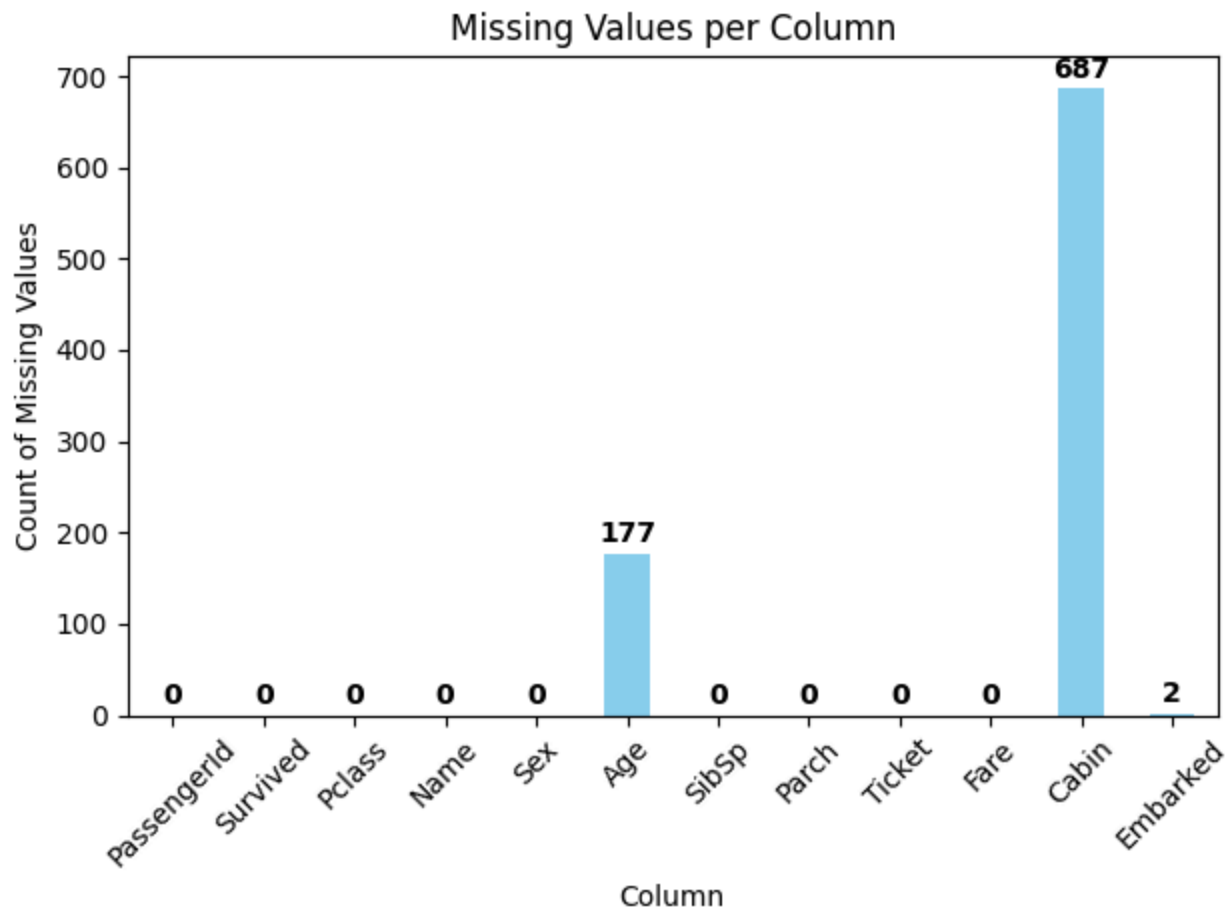
```

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0

```
Cabin          687  
Embarked       2  
dtype: int64
```

```
missing = df.isnull().sum()
```

```
# Create the bar plot  
ax = missing.plot(kind='bar', color='skyblue')  
plt.title('Missing Values per Column')  
plt.ylabel('Count of Missing Values')  
plt.xlabel('Column')  
plt.xticks(rotation=45)  
# Add text labels above each bar  
for index, value in enumerate(missing):  
    plt.text(index, value+5, str(value), ha='center', va='bottom', fontsize=10, fontweight='bold')  
plt.tight_layout()  
plt.show()
```



```
df = df.drop(columns=['Cabin']) # Cabin has 687 missing – too many!
```

```
df['Age'] = df['Age'].fillna(df['Age'].median())
```

```
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
```

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	0
Embarked	0

dtype: int64

```
# Count the number of people who survived and didn't
survival_counts = df['Survived'].value_counts()
print("Survival Counts:\n", survival_counts)
```

```
Survival Counts:
Survived
0      549
1      342
Name: count, dtype: int64
```

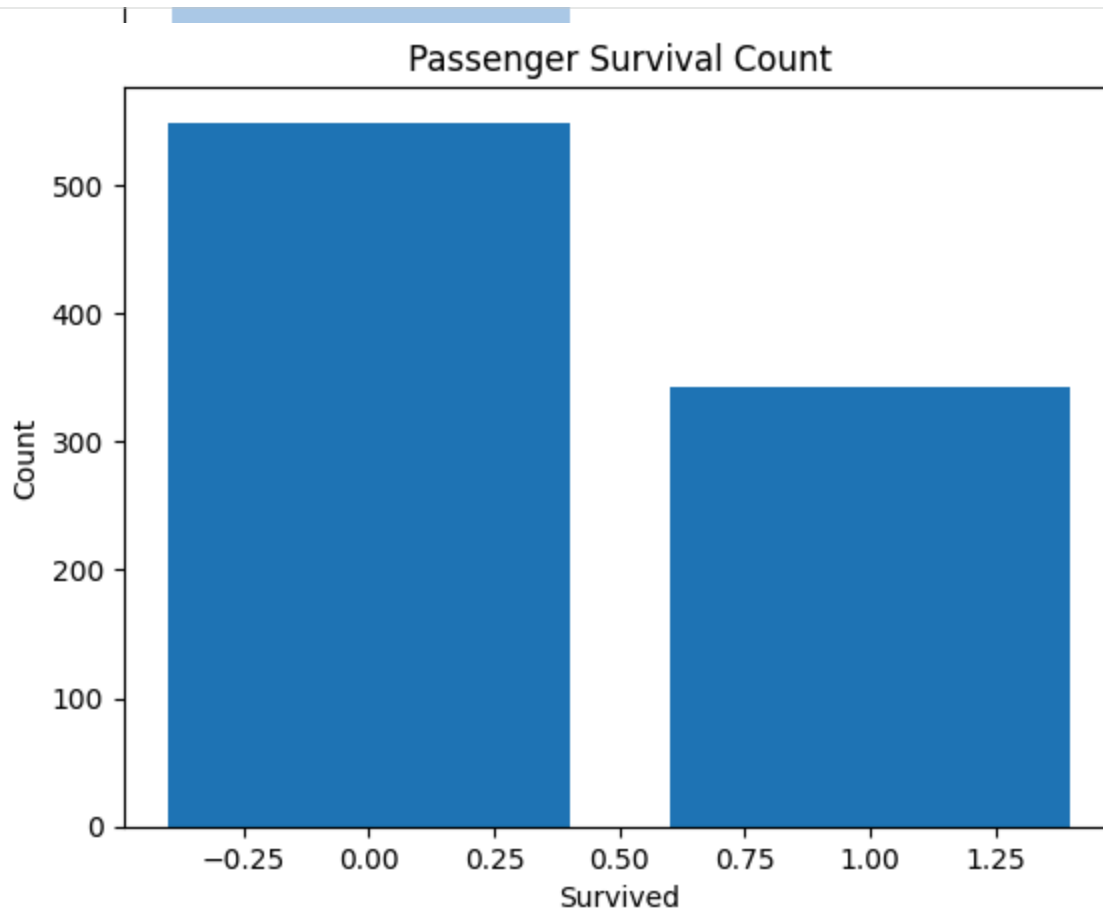
```
#Survival Count (Bar Plot)
sns.countplot(x='Survived', data=df, palette='pastel')
```

```
plt.rcParams['text.usetex'] = True
plt.title('Survival Count (0 = Died, 1 = Survived)')
plt.xlabel('Survived')
plt.ylabel('Passenger Count')
plt.show()
```

```
/tmp/ipykernel_4163/927445151.py:1: FutureWarning:
```

```
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign t
```

```
counts = df['Survived'].value_counts() # counts of 0 and 1
plt.bar(counts.index, counts.values)
plt.xlabel("Survived")
plt.ylabel("Count")
plt.title("Passenger Survival Count")
plt.show()
```



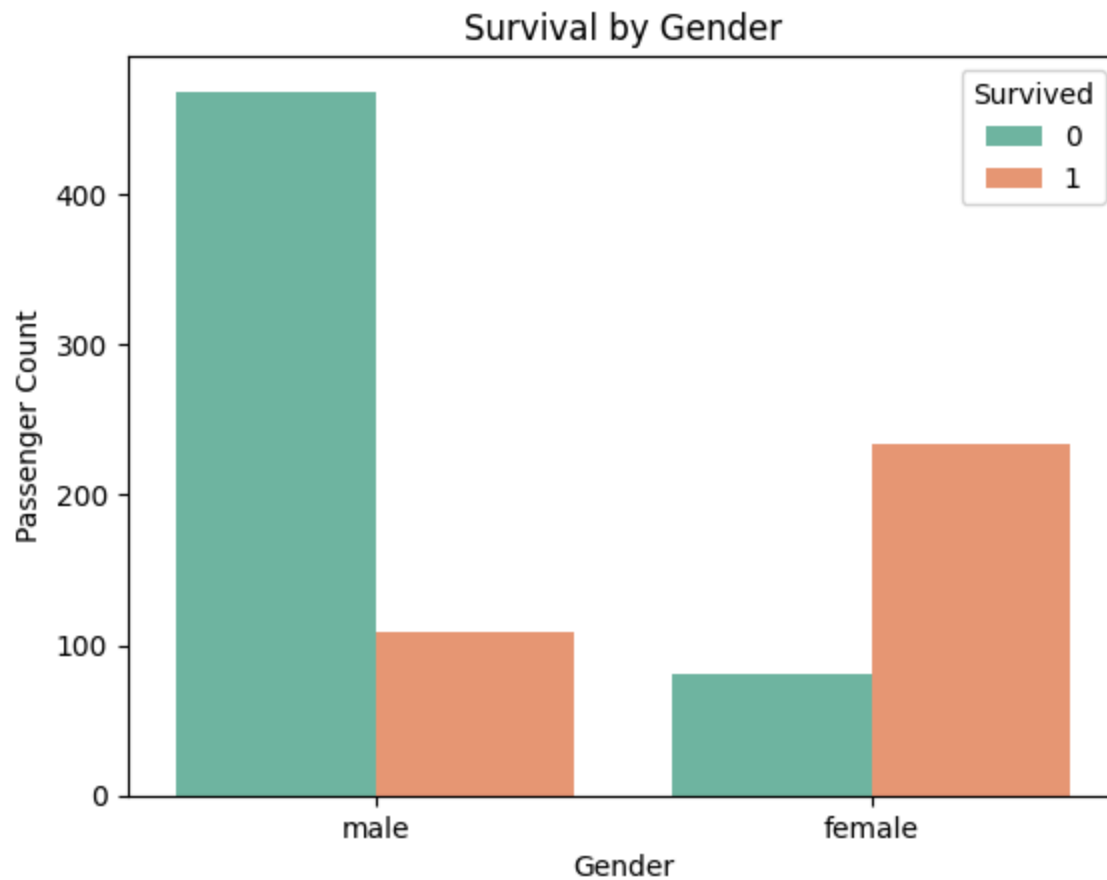
```
# Count of passengers by gender
gender_counts = df['Sex'].value_counts()
print("\nGender Counts:\n", gender_counts)
```

```
Gender Counts:
Sex
male      577
female    314
Name: count, dtype: int64
```

```
# Count of survived grouped by gender
survival_by_gender = df.groupby(['Sex', 'Survived']).size()
print("\nSurvival by Gender:\n", survival_by_gender)
```

```
Survival by Gender:
Sex      Survived
female  0          81
        1         233
male    0         468
        1         109
dtype: int64
```

```
#Survival by Gender
sns.countplot(x='Sex', hue='Survived', data=df, palette='Set2')
plt.title('Survival by Gender')
plt.xlabel('Gender')
plt.ylabel('Passenger Count')
plt.legend(title='Survived')
plt.show()
```



```
#Passenger Class Distribution
# Count of passengers by class
pclass_counts = df['Pclass'].value_counts()
print("\nPassenger Class Counts:\n", pclass_counts)
```

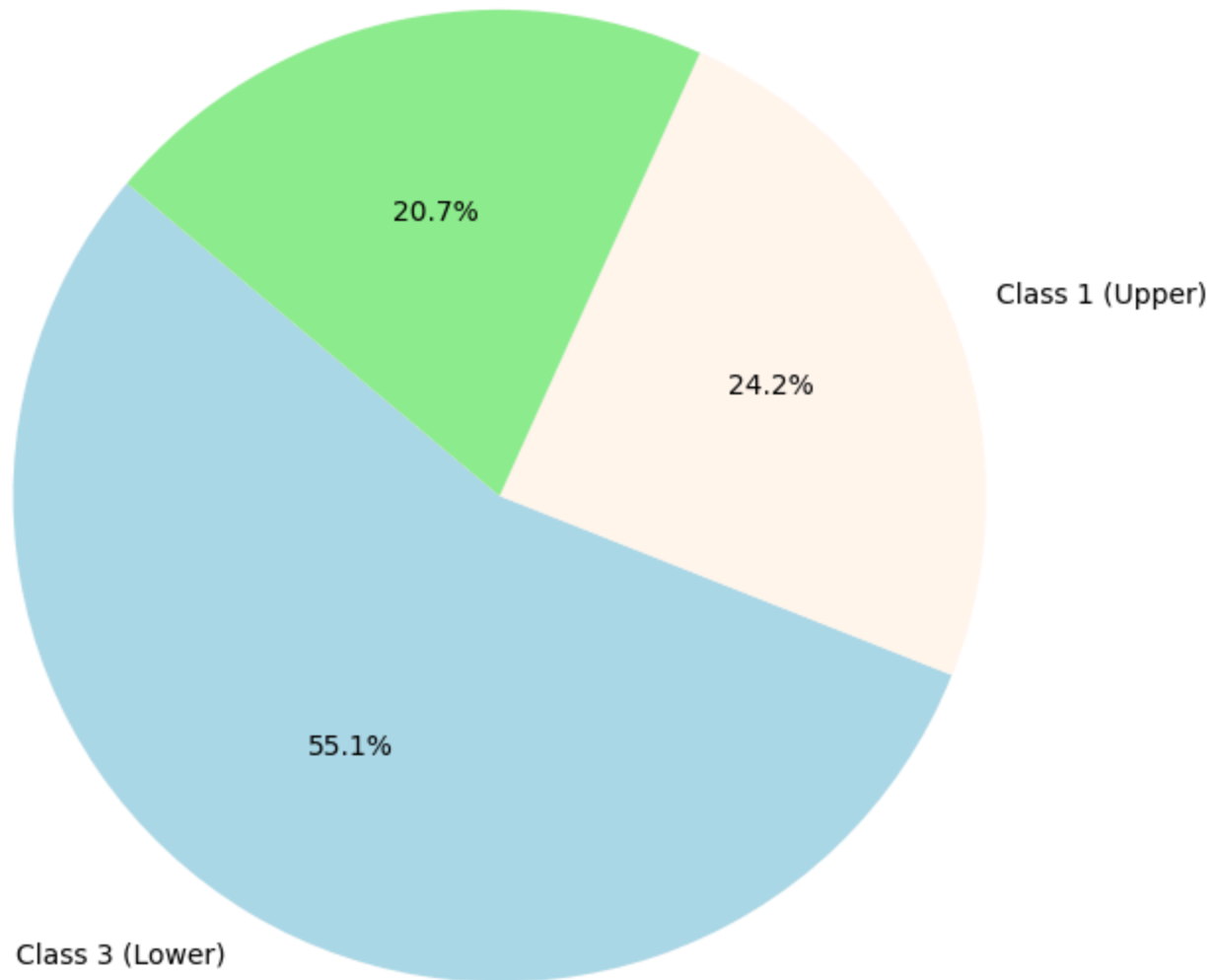
```
Passenger Class Counts:
Pclass
3      491
1      216
```

2 184

Name: count, dtype: int64

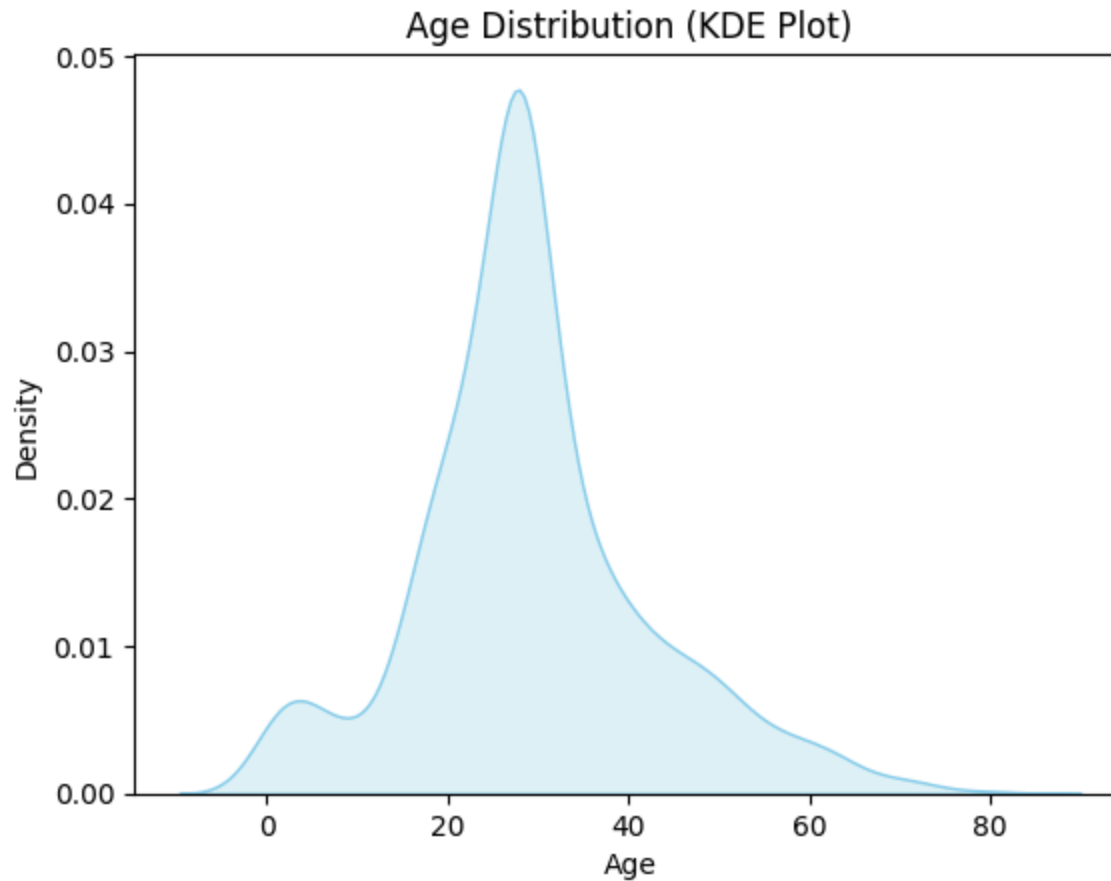
```
#Passenger Class Distribution
# Labels for the pie chart
labels = ['Class 3 (Lower)', 'Class 1 (Upper)', 'Class 2 (Middle)']
# Plotting the pie chart
plt.figure(figsize=(7, 7))
plt.pie(pclass_counts, labels=labels, autopct='%1.1f%%', startangle=140, colors=['lightblue', 'lightcoral', 'lightgreen'])
plt.title('Passenger Class Distribution (Pclass)', fontsize=14)
plt.axis('equal') # Ensures it's a circle
plt.show()
```

Passenger Class Distribution (Pclass)
Class 2 (Middle)



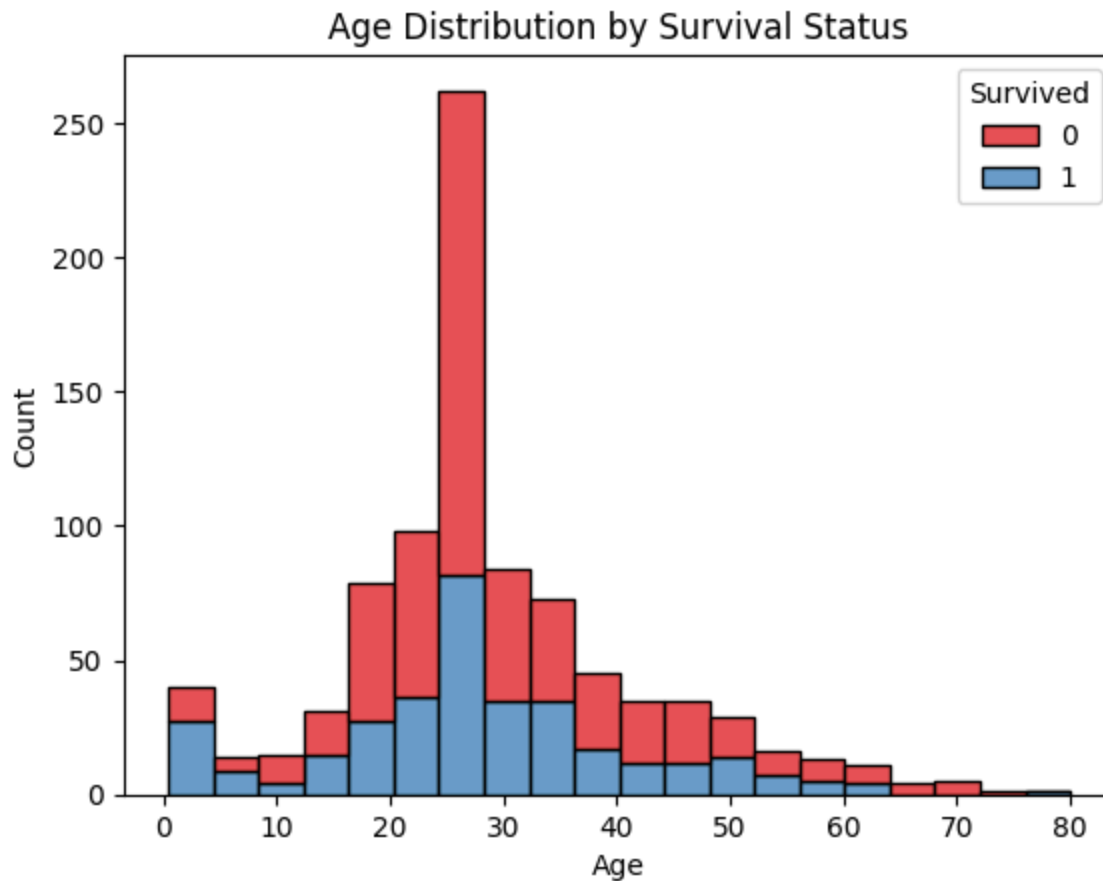
```
#Age Distribution ( KDE Plot (Smooth Curve Distribution  
sns.kdeplot(df['Age'], fill=True, color='skyblue')  
plt.title('Age Distribution (KDE Plot)')
```

```
plt.xlabel('Age')  
plt.ylabel('Density')  
plt.show()
```



#Age vs Survival

```
sns.histplot(data=df, x='Age', hue='Survived', multiple='stack', bins=20, palette='Set1')  
plt.title('Age Distribution by Survival Status')  
plt.xlabel('Age')
```



```
#survival by age group
df['AgeGroup'] = pd.cut(df['Age'], bins=[0, 12, 18, 40, 60, 100], labels=['Child', 'Teen', 'Adu
sns.countplot(x='AgeGroup', hue='Survived', data=df)
plt.title('Survival by Age Group')
plt.show()
```